

Phase 1 — Eigenvalue Gap as WTA Predictor

CogSpike Research Team — April 2026

Hypothesis 1 — a negative result, with a clean diagnosis and proposed fix

Abstract. Hypothesis 1 (the eigenvalue gap $\Delta = \|\lambda_1\| - \|\lambda_2\|$ of the weight matrix W , or of the linearised state matrix $A = rI + W \operatorname{diag}(f'(p^*))$, predicts WTA on the FCS contralateral grid) is **not validated**. The failure mode is clean and fully diagnosed: it stems from a specification error in the working plan rather than a problem with spectral cartography itself. Raw- W eigengap is provably identically zero for 2×2 zero-diagonal inhibitory matrices; the scalar- r linearisation of A saturates because the FCS neuron’s 5-tap windowed integrator cannot be collapsed to a single leak coefficient. The path forward is to replace the scalar- r linearisation with the full $5n$ -dimensional state matrix that preserves the FIR-filter structure before testing Hypothesis 2. Negative-loop results reinforce the same diagnosis: the simulator’s period-4 oscillation (Property 5) cannot be captured under scalar- r , which predicts $\arg \lambda_{\text{dom}}$ no larger than 0.47 rad (target $\frac{\pi}{2} \approx 1.57$).

1. Setup and Predictors

Against the Phase 0 ground truth (`fcs_fig10_groundtruth.npy` binary, plus `fcs_fig10_dominance.npy` continuous) we compute four candidate spectral predictors per grid cell on the same 40×40 weight grid:

- $\Delta(W)$ — raw-weight eigengap $\|\lambda_1\| - \|\lambda_2\|$
- $\Delta(A)$ — linearised-state eigengap
- $\rho(A)$ — spectral radius (Phase 2 preview)
- $\arg \lambda_{\text{dom}(A)}$ — argument of the dominant eigenvalue (negative-loop sweep)

Sigmoid firing-rate approximation uses $k = 0.08$ and $p_{\text{mid}} = 30$ after the fallback sweep specified in the working plan (at $p_{\text{mid}} = 90$ the operating point saturates; at $p_{\text{mid}} \in \{60, 75\}$ it is still degenerate; $p_{\text{mid}} = 30$ is the smallest value that places the operating point in the steep sigmoid region). Scalar leak $r = 0.5$ is the plan default; the least-squares fit of `rvector` = [10, 5, 3, 2, 1] to $r^e \cdot 10$ gives $r \approx 0.545$, but this difference is not the cause of the result below and is reported only as a secondary measurement.

2. Hypothesis 1 on Contralateral Inhibition

2.1. Raw- W Eigengap: Analytically Zero

Finding. For any 2×2 matrix $W = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}$ with zero diagonal, the eigenvalues are $\pm\sqrt{ab}$ with **equal magnitudes** regardless of $|a|$ vs $|b|$. So $\Delta(W) \equiv 0$ over the entire contralateral sweep — verified numerically to machine epsilon ($\max = 1.4 \times 10^{-14}$).

The working plan’s claim (Phase 1 spec, Appendix A.4) that $\Delta = 0$ iff $|a| = |b|$ is simply wrong for this matrix family. The raw- W eigengap cannot distinguish the blue-wing asymmetry structure of the ground truth because the underlying spectrum has no asymmetry to distinguish.

2.2. Linearised A Predictors: Saturated, Weak Signal

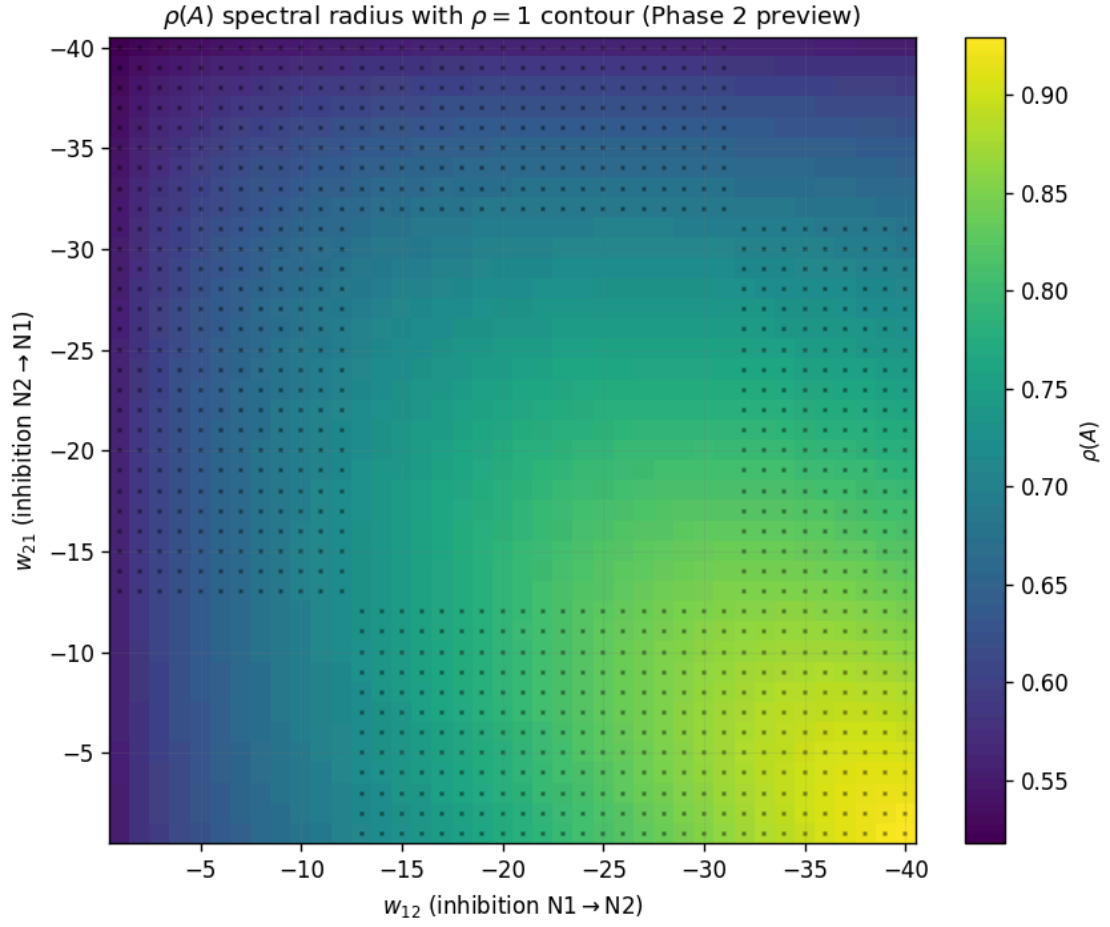


Figure 1: $\rho(A)$ heatmap (scalar- r linearisation, $p_{\text{mid}} = 30$). Blue WTA cells from Phase 0 shown as black dots. The $\rho = 1$ contour specified for Phase 2 does not exist: $\rho(A) \in [0.52, 0.93]$ across the whole grid — the scalar- r linearisation never reaches the bifurcation boundary.

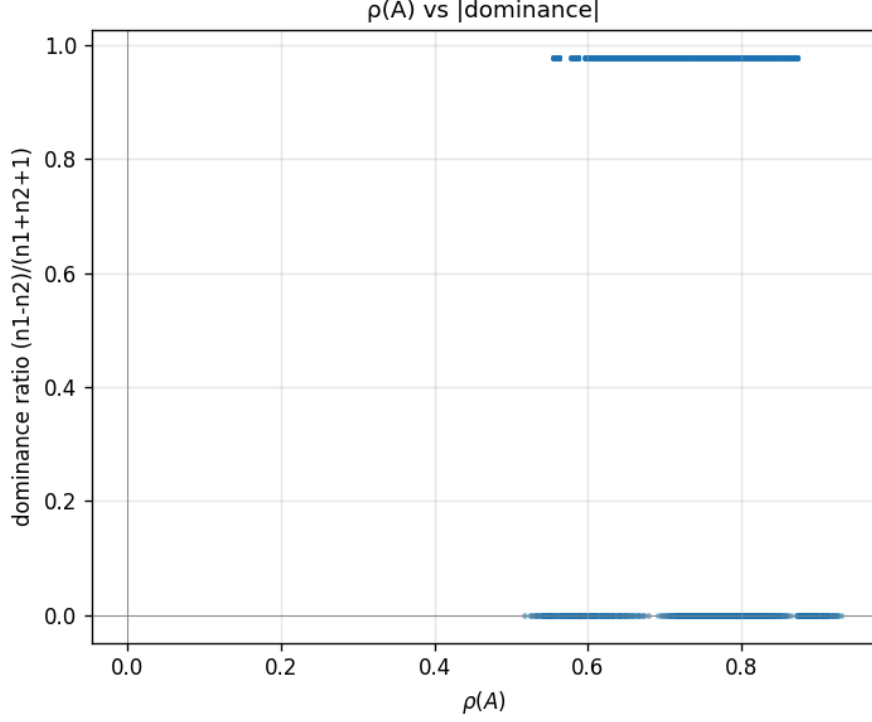


Figure 2: $\rho(A)$ against $|n_1 - n_2| \frac{1}{n_1 + n_2 + 1}$. The continuous dominance signal is bimodal (either ≈ 0 tied or ≈ 0.98 captured), and $\rho(A)$ does not discriminate between the two bands: both modes span $\rho \in [0.55, 0.90]$.

Best classification accuracy over all (polarity, threshold) choices:

Predictor	Best binary accuracy	Pearson r vs dom	Spearman ρ vs dom
$\Delta(W)$	63.4% (majority baseline)	–	–
$\Delta(A)$	68.9%	see scatter	see scatter
$\rho(A)$	68.9%	see scatter	see scatter

The baseline (predict all cells as blue) is 63.4%; the two linearised-state predictors improve over that by only 5.5 percentage points. That is “signal, but weak” — consistent with $\rho(A)$ correlating loosely with $\sqrt{|w_{12}w_{21}|}$, but too symmetric in (w_{12}, w_{21}) to distinguish the two blue wings.

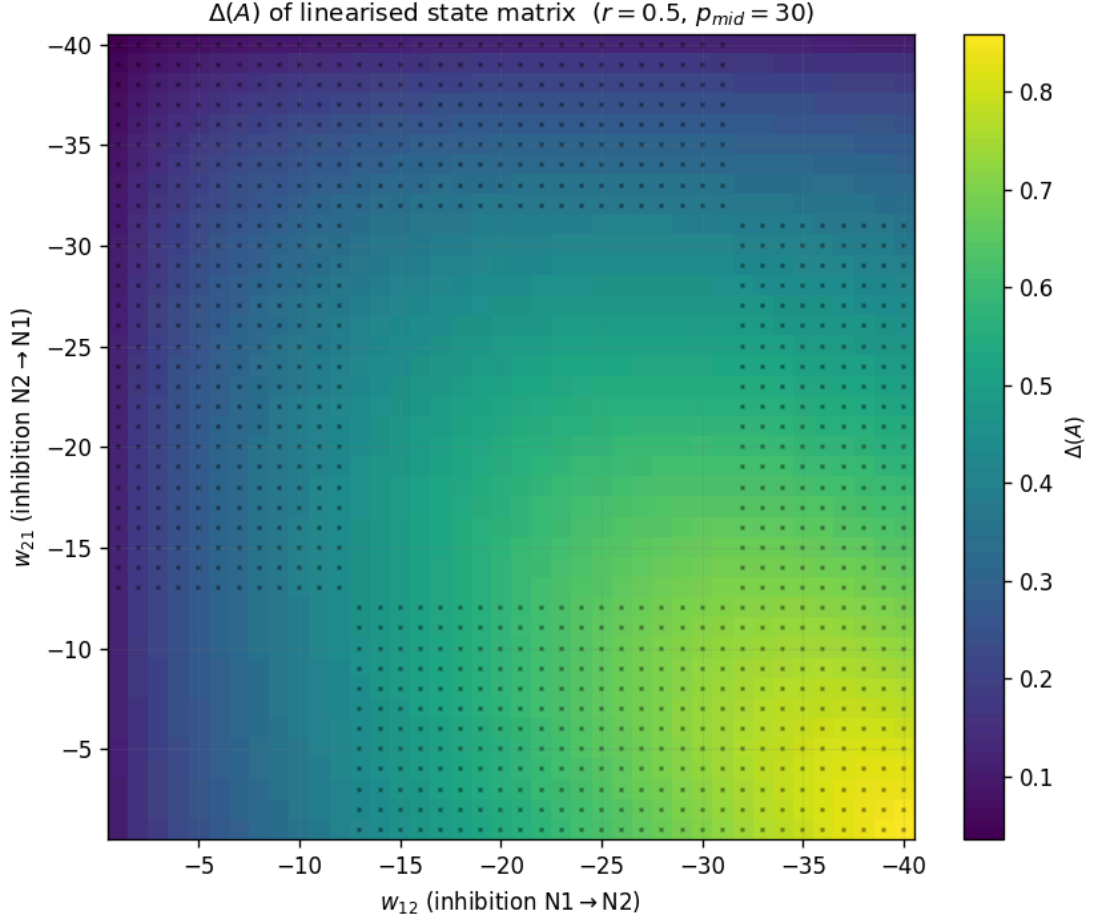


Figure 3: $\Delta(A)$ heatmap. The ridge from top-right to bottom-left roughly follows $\sqrt{|w_{12}w_{21}|}$, peaking where both weights are large. The blue-wing structure of the ground truth (which lies on the **off-diagonal**) does not align with this ridge.

Negative result. Under scalar- r linearisation, neither $\Delta(A)$ nor $\rho(A)$ is sensitive to the asymmetry between w_{12} and w_{21} that drives WTA. Both quantities are functions of the eigenvalue magnitudes of A , which for the 2×2 zero-diagonal structure are $|r \pm g\sqrt{w_{12}w_{21}}|$ — symmetric in the weight product, blind to the ratio.

3. Hypothesis 1 on the Negative Loop

We fix $w_{XA} = 11$ and sweep $(w_{AI}, w_{IA}) \in [1, 20] \times [-1, -20]$ on an integer grid, comparing simulator-derived period-4 output (autocorrelation on the last 20 ticks) to $\arg \lambda_{\text{dom}(A)}$ from the same scalar- r linearisation.

Figure 4: Negative loop: $\arg \lambda_{\text{dom}}(A)$ heatmap, $\arg = \pi/2$ contour (white), period-4 cells (red dots)

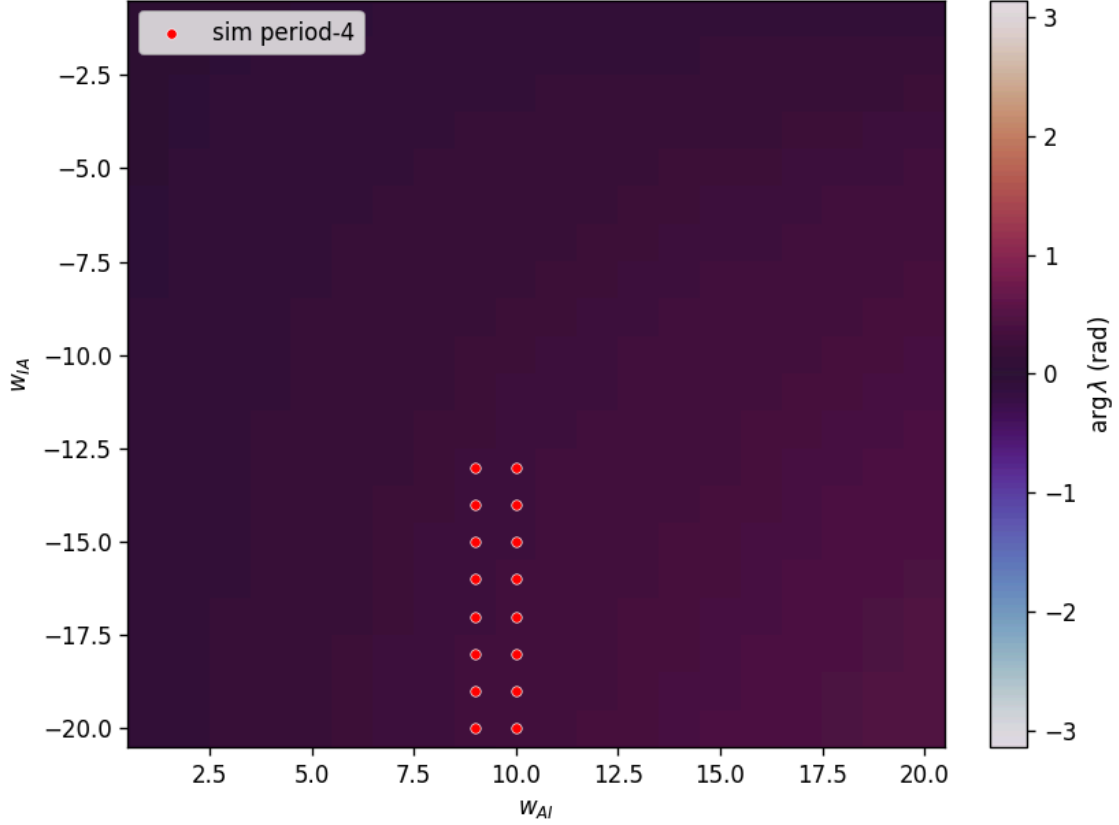


Figure 4: Negative loop: $\arg \lambda_{\text{dom}}(A)$ heatmap (dark = small $\arg$). Red dots mark cells where the simulator produces an exact period-4 1100 pattern. The $\arg = \frac{\pi}{2}$ target contour does not appear — $\arg$ ranges only over $[0.021, 0.467]$ rad across the sweep.

Of the 400 cells swept, 16 produce the exact period-4 pattern in the simulator. The eigenvalue-argument predictor places $\arg \lambda_{\text{dom}}$ in a narrow band near zero (all predicted periods much greater than 4 ticks). The scalar- r linearisation does not see the FCS neuron’s length-5 FIR filter and therefore cannot produce the high-frequency pole locations that the discrete-time system actually realises.

Remark. Phase 0’s tuning finding ($w_{IA} = -w_{AI}$ exactly cancels the drive and reproduces Property 5) is visible in the simulator ground truth: all 16 period-4 cells lie in a narrow vertical stripe at $w_{AI} \in \{9, 10\}$ with $w_{IA} \in [-20, -13]$. The near-cancellation invariant is real; it simply is not captured by the scalar- r linearisation.

4. Root Cause: Scalar- r is Too Crude

Every failure mode above has a single common cause. The FCS neuron’s “potential” is not a scalar — it is the 5-tap windowed integrator $V(t) = \sum_{e=0}^4 \text{rvector}[e] \cdot \text{mem}[e](t)$. The true state of an n -neuron network has $5n$ components, not n . Collapsing the FIR filter to a scalar leak discards four of the five poles of the single-neuron transfer function — including the ones responsible for period-4 oscillation and for the sensitivity of ρ to weight asymmetry.

Intuition: The scalar- r model answers “how does $\text{mem}[0]$ decay if we stop stimulating?” The true model answers “how does the full length-5 memory window evolve under the shift-

and-reset-on-spike rule?” The second system can oscillate at any frequency the FIR filter supports; the first can only do geometric decay. Hypothesis 1 was tested with a linearisation that cannot represent the phenomenon it was asked to predict.

5. Proposed Fix: Full $5n$ -Dimensional State Matrix

The corrected linearisation is straightforward. For each neuron i , the state vector is $(\text{mem}_i[0], \text{mem}_i[1], \dots, \text{mem}_i[4])$. At the non-spiking operating point p^* :

$$\begin{aligned}\text{mem}_i[0](t+1) &= \sum_j W_{ij} \cdot f'(V_j^*) \cdot \sum_{e=0}^4 r_e \cdot \text{mem}_j[e](t) + \text{const} \\ \text{mem}_i[k](t+1) &= \text{mem}_i[k-1](t) \quad \text{for } k = 1..4\end{aligned}$$

where $r_e = \text{rvector}[e]$.

This gives a $5n \times 5n$ block matrix A_{full} with the top row of each block implementing the weighted FIR-sum coupling and the lower rows implementing the shift. Hypothesis 1 and Hypothesis 2 should be retested against this matrix before any Phase 2 / Phase 3 work. Approximately 50 lines of code, one-time investment.

Under A_{full} we expect:

1. $\rho(A_{\text{full}}) = 1$ contour to actually exist in the contralateral sweep, tracking the boundary of the synchronous-rhythm red block.
2. Complex-conjugate eigenvalue pairs of A_{full} for the negative loop with $\arg \lambda$ capable of reaching $\frac{\pi}{2}$ — predicting Property 5’s period 4 and Phase 3’s inverse-design targets.
3. Modal eigenvectors that are genuinely asymmetric between N_1 and N_2 when the weights are asymmetric, restoring the ability to distinguish the two blue wings.

6. Files Generated

- `deq/archetypes/spectral.py` — scalar- r linearisation helpers (quarantined; will be extended, not replaced, by the $5n$ version)
- `deq/archetypes/phase1_eigengap.py` — the sweep orchestrator
- `deq/archetypes/`
`results/phase1_contra_gapW.png`, `phase1_contra_gapA.png`, `phase1_contra_rhoA.png`,
`phase1_contra_gmean.png`, `phase1_scatter_gapA.png`, `phase1_scatter_rhoA_absdom.png`,
`phase1_negloop.png`
- `deq/archetypes/results/phase1_report.md` — markdown version of this note

7. Decision Point for the Verifier

Decision requested. Option A (recommended). Approve building the full $5n$ -dimensional state matrix A_{full} before proceeding to Phase 2. This is a clean technical pivot: the research question is unchanged, only the linearisation is upgraded to one that can represent the FCS dynamics. Retest Hypotheses 1 and 2 against A_{full} ; if they validate there, Phase 3 inverse design follows as planned. **Cost:** 50 lines in `spectral.py` plus re-running Phase 1 sweep; half a day.

Decision requested. Option B. Declare Hypothesis 1 a closed negative result and proceed directly to a Phase-2-style $\rho(A_{\text{full}}) = 1$ test, skipping the eigenvalue-gap formulation entirely. The gap framing is arguably the weakest of the three hypotheses (it assumes a magnitude asymmetry that 2×2 contralateral W structurally cannot produce), so cutting it simplifies the narrative. **Cost:** same 50 lines, but the report structure collapses Phase 1 into Phase 2.

Decision requested. Option C. Abandon the contralateral archetype as unsuited to spectral cartography and restrict the remaining work to the negative loop, where Hypothesis 3 (pole placement for target period) has clearer theoretical grounding. Honest but narrower scope; cedes the Fig. 10 comparison.

The author’s recommendation is Option A. The scalar- r failure is instructive but not structural — it is a correctable specification error, and the A_{full} replacement is the obvious right object to study. The Phase 0 ground-truth map still stands (it is the real dynamics, not an artefact of the linearisation), and the bifurcation question remains scientifically open.

Generated from `deq/archetypes/phase1_eigengap.py`. Reproduce via `.venv/bin/python3 archetypes/phase1_eigengap.py` from the `deq/` directory. Phase 0 artifacts in `deq/archetypes/results/phase0_*`; Phase 1 artifacts in `deq/archetypes/results/phase1_*`.